

Reilly Raab

Data Scientist — Level 2

Pacific Northwest National Laboratory

Remote, USA
Nov 2025 – Present

Safety, Security, and Responsible Use of Agentic AI.

- Technical lead for “CommentNEPA”.
 - One of three PermitAI applications funded by Permitting Council for use by federal agencies.
 - Under evaluation by DOE, NPS, BLM, NRC, USDA, NIST, et. al.
- Safety and explainability team for DOE’s “Project Genesis”.

Postdoctoral Research Associate

Pacific Northwest National Laboratory

Remote, USA
Oct 2024 – Nov 2025

Self-optimizing AI workflows and online, physics-informed control.

- Technical lead for “CommentNEPA”,
 - Auditable, AI-enabled data analysis for public comment processing
 - 1st author, arXiv preprint (2025): “*Audit, Alignment, and Optimization of LM-Powered Subroutines with Application to Public Comment Processing*”.

Graduate Student Researcher (PhD)

UC Santa Cruz

Santa Cruz, CA
Sept 2019 – Mar 2024

Doctoral research developing safe AI/ML systems for high-stakes decisions while accounting for dynamical feedback.

- Dissertation: “*Machine Learning and the Multiagent Alignment Problem*”.
- Applied natural gradient descent to model adaptive populations according to evolutionary game theory.
 - 1st author, **NeurIPS (2021)**: “*Unintended Selection: Persistent Qualification Rate Disparities and Interventions*”.
 - **Spotlight Paper Award** • Top 50 review score of ~ 9100 submissions
 - 1st author, arXiv preprint (2023): “*Conjugate Natural Selection*”.
- Developed constraint-violation bounds for ML systems subject to adversarial distribution shift.
 - 1st co-author, **NeurIPS (2022)**: “*Fairness Transferability Subject to Bounded Distribution Shift*”.
- Established novel theoretical safety guarantees using online reinforcement learning.
 - 1st co-author, **NeurIPS (2023)**: “*Long-Term Fairness with Unknown Dynamics*”.
 - Highlighted Paper and Best Paper Runner-Up: ICLR RTML Workshop (2022)
- Formulated constrained optimization programs to sequentially adapt ML policies.
 - 1st author, **AAAI (2024)**: “*Fair Participation via Sequential Policies*”.

Software Developer

Breadware, Inc.

Reno, NV
Oct 2016 – Aug 2018

Early startup offering rapid prototyping services for the internet-of-things (IoT).

- Automated electronic design tasks, circuit board layout (Python).
- Architected a system to generate netlists and firmware from modular specification (Python, C).
- Implemented web-based testing of user logic for embedded devices in simulated environments (JavaScript).

EDUCATION

PhD, Computer Science and Engineering

University of California, Santa Cruz

Santa Cruz, CA
Sept 2019 – Mar 2024

- ARCS Scholar (Northern California) • Dean’s Fellow • Regents Fellow • Dissertation Year Fellow

BSc, Physics

University of California, Santa Barbara (College of Creative Studies)

Santa Barbara, CA
Sept 2011 – June 2015

- Distinction in the Major • High Honors • Multiple Education Abroad Scholarships

SKILLS

Domains: ML/RL, Game Theory, Constrained Optimization, Scientific Computing, Agentic AI.
Languages (Proficient): C, Python (Scientific Computing, ML/RL, Web Backends, etc.), TS / JS.
Tooling: GNU/Linux, Slurm, Git, Docker, Cloud, SQL